ME 101 BME

Model questions on I C Engine portion(MOD-4)

1. Define engine?
2. Distinguish between SI and CI engine.
3. Enumerate the various parts of an I C engine.
4. Explain the working of 2 Stroke petrol engine.
5. Explain the working of 4 Stroke petrol engine.
6. Explain the working of 4 Stroke diesel engine.
7. What are the merits and demerits of 2 Stroke engine?
8. What are the merits and demerits of 4 Stroke engine?
9. Compare petrol engine with diesel engine.
10. Discuss the latest trends in engines used for cars.

Module 4: Boilers and Internal Combustion Engine:

Boiler Mountings and Accessories, Fire Tube and Water Tube Boilers, Cochran Boiler, Babcock and Wilcox Boiler.

Model questions on Boilers

9.1. State how the boilers are classified.

9.2. Explain the principle of fire tube and water tube boilers.

9.3. Describe with a neat sketch the working of Cochran boiler. Show the position of different mountings and explain the function of each.

9.4. Describe, giving neat sketches, the construction and working of a Lancashire boiler. Show the positions of different mountings and accessories.

9.5. Sketch and describe the working of a Locomotive boiler. Show the positions of fusible plug, blow off cock, feed check valve and superheater. Mention the function of each. Describe the method of obtaining draught in this boiler.

9.6. Give an outline sketch showing the arrangement of water tubes and furnace of a Babcock and Wilcox boiler. Indicate on it, the path of the flue gases and water circulation. Show the positions of fusible plug, blow off cock and superheater. Mention the function of each.

9.7. Explain why the superheater tubes are flooded with water at the starting of the boilers ?

9.8. Mention the chief advantages and disadvantages of fire tube boilers over water tube boilers.

9.9. Discuss the chief advantages of water tube boilers over fire tube boilers.

9.10. What are the considerations which would guide you in selecting the type of boiler to be adopted for a specific purpose ?

9.11. Distinguish between water-tube and fire-tube boilers and state under what circumstances each type would be desirable.

Model questions on Boiler Mountings and Accessories

10.1. Describe with a neat sketch water level indicator and explain how the flow of steam and water is automatically stopped when the glass tube breaks.

10.2. Explain why safety valves are needed in a boiler. Draw a neat sketch of spring-loaded safety valve and explain its working.

10.3. What do you understand by high steam, low water safety valve ? Explain its working with a neat sketch.

10.4. What are the major differences between mountings and accessories ? Give three examples of each.

10.5. Draw a neat sketch of a Bourdon pressure gauge and explain its working principle.

10.6. What is the purpose of steam stop valve ? Explain its working.

10.7. What is the purpose of a fusible plug ? Explain its working. Indicate its usual location for Cochran, Lancashire, Locomotive and Babcock and Wilcox boilers.

10.8. Explain why the blow-off cock is operated periodically when the boiler is working. Where is it located ? Explain its working with a neat sketch.

10.9. What do you understand by feed check valve ? Explain the working of a feed check valve with a neat sketch.

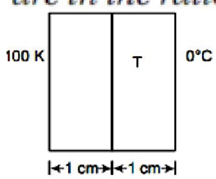
10.10. Draw a neat diagram of Green Economiser and explain its working.

10.11. What are the advantages of preheating the air ? Draw a neat diagram of tubular air-preheater and explain its working.

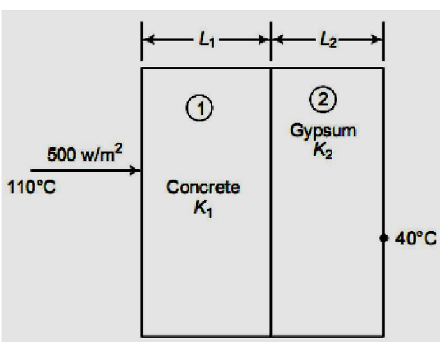
10.12. Explain the functions of a superheater. With a neat sketch, describe the working of a superheater used in Lancashire Boilers.

Q1. *The inside surface of a 25cm thick brick wall* $(K = 0.69 W/mK)$ *is at* $27°C$ *while the outside surface is at* $2°C$. *Determine the rate of heat loss per unit area through the brick wall.*

Q2. *Sheets of brass and steel, each of thickness 1 cm, are placed in contact. The outer surface of brass is kept at* $100°C$ *and outer surface of steel is kept of* $0°C$. *What is the temperature of the common interface? The thermal conductivities of brass and steel are in the ratio of* $2:1$.



Q3. *A concrete wall* $(K = 1.37 W/mk)$ *of* $10\ cm$ *thickness is to be plastered with gypsum* $(K = 0.48 W/mk)$, *so that the heat losses from the wall do not exceed* $500 W/mk$ *when the inner and outer surfaces of the wall are at* $110°C$ *and* $40°C$, *respectively. Determine the thickness of plastering to be added to the concrete wall.* $A = 1\ m^2$



Q4. Find the rate of heat transfer per meter length of the compost wall.